

ADVANCED DATA VISUALIZATION PROJECT REPORT

(Project Semester Mar-June 2026)

Job Market Analytics Dashboard Software Hiring Trends in India

Submitted by

Mannuru Mahesh

Registration No.

12504689

Programme and Section- M.Tech Data Science & Analytics (325SD)

Course Code- INT555



Under the Guidance of komal Arora

School of Computing and AI

Lovely School of Computer Science and Engineering

DECLARATION

I, Mannuru Mahesh, student of M.Tech Data Science & Analytics under School of Computing and AI at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date:

Signature

Registration No. 12504689

Mannuru Mahesh

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify Mannuru Mahesh, bearing Registration no. 12504689 has completed Advanced Data Visualization (INT555) project titled, “**Job Market Analytics Dashboard Software Hiring trends in india**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor: Komal Arora

Designation of the Supervisor: Professor

School of Computing and AI

Lovely Professional University Phagwara, Punjab

Date:

ACKNOWLEDGEMENT

The successful completion of this project would not have been possible without the support, guidance, and encouragement of many individuals. We feel truly privileged to have received their assistance throughout the journey of this work.

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I am also grateful to all the faculty members of **Lovely Professional University** for their continuous support, motivation, and technical guidance. Their expertise and willingness to help played a significant role in shaping the successful outcome of this project.

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Abstract

The Job Market Analytics Dashboard Software Hiring Trends in India is designed to provide a comprehensive analysis of the current recruitment landscape in the software industry across various Indian cities. The dashboard visualizes important hiring metrics such as total job openings, top hiring companies, salary distributions, experience-level demands, and location-based opportunities. By integrating interactive charts, maps, and KPI indicators, the dashboard helps users easily explore and understand large volumes of recruitment data.

This dashboard mainly focuses on identifying software hiring trends by analyzing job postings collected from different organizations and recruitment platforms. It highlights the most in-demand job roles, frequently hiring companies, and preferred experience levels in the Indian IT sector. The analysis also provides insights into salary ranges offered for different technical roles, enabling users to understand compensation trends in the market. These insights are valuable for job seekers, recruiters, and business analysts who want to make informed decisions based on real-time hiring patterns.

The project uses data visualization and business intelligence techniques to transform raw recruitment data into meaningful and actionable insights. Various Power BI visual elements such as bar charts, donut charts, treemaps, KPI cards, and geographical maps are used to present the information in an interactive and user-friendly manner. Filters and slicers are implemented to allow users to customize the analysis based on city, job title, experience band, and salary category. This enhances the usability and flexibility of the dashboard for detailed market exploration.

One of the key objectives of this dashboard is to help aspiring professionals and students understand the evolving demands of the software job market in India. By analyzing hiring trends and skill requirements, users can identify the most promising career opportunities and prepare themselves according to industry expectations. Similarly, recruiters and HR professionals can utilize the dashboard to monitor hiring competition, talent availability, and regional recruitment trends to improve workforce planning and recruitment strategies.

Overall, the Job Market Analytics Dashboard demonstrates the effective use of data analytics and visualization in solving real-world business problems related to recruitment and employment analysis. The dashboard provides a centralized platform for monitoring software hiring activities and supports data-driven decision-making through clear, attractive, and interactive visual representations. This project showcases the practical application of Power BI, data cleaning, data transformation, and analytical storytelling in the field of job market intelligence.

Introduction

The rapid growth of the Information Technology (IT) industry has significantly transformed the employment landscape in India. With the increasing demand for software professionals, organizations are continuously hiring skilled candidates for various technical roles such as Data Analyst, Software Engineer, Data Scientist, Web Developer, and Cloud Engineer. Analyzing these hiring patterns has become essential for understanding current job market trends and identifying future employment opportunities.

The Job Market Analytics Dashboard Software Hiring Trends in India is developed to provide a detailed and interactive analysis of software job openings across different cities and companies in India. The dashboard helps users visualize recruitment trends, salary distributions, experience-level requirements, and top hiring organizations through meaningful charts and visualizations. It acts as a centralized platform for monitoring and exploring recruitment-related data effectively.

The main purpose of this project is to transform raw job posting data into actionable insights using data analytics and business intelligence techniques. Large volumes of recruitment data often contain hidden patterns and trends that are difficult to understand in their raw form. By applying data cleaning, transformation, and visualization techniques, the dashboard simplifies complex datasets into easy-to-understand graphical representations.

This dashboard is built using Power BI, which enables the creation of dynamic and interactive visualizations. Different visual elements such as KPI cards, treemaps, donut charts, bar charts, line graphs, and geographical maps are incorporated to represent the hiring trends in a visually appealing manner. Interactive filters and slicers are also included to allow users to customize the analysis based on city, job title, salary range, and experience level.

One of the important features of this project is the analysis of salary trends in the software industry. The dashboard provides insights into average salary packages, salary bands, and compensation distribution for different job roles and experience levels. This information helps job seekers understand market standards and assists companies in designing competitive salary packages.

The dashboard also highlights the top hiring companies and most in-demand job roles in India. By analyzing the number of job openings across organizations, users can identify companies with active recruitment processes and understand the technologies and skills that are currently trending in the software market. This enables students and professionals to focus on the most relevant skills for career growth.

Another significant aspect of the project is experience-level analysis. The dashboard categorizes job openings into different experience bands such as fresher, junior-level, mid-level, and senior-level positions. This helps users identify the availability of opportunities according to their experience and allows recruiters to understand workforce demand across different professional levels.

Geographical analysis is also an important component of the dashboard. The project visualizes job openings

across major Indian cities such as Bengaluru, Hyderabad, Pune, Chennai, Mumbai, and Delhi. This enables users to identify cities with high recruitment activity and better employment opportunities in the software industry. The map visualizations provide a clear understanding of regional hiring distributions.

The project demonstrates the practical application of data analytics concepts including data cleaning, data preprocessing, data transformation, exploratory data analysis (EDA), and dashboard development. DAX calculations and Power Query transformations are used to enhance the analytical capabilities of the dashboard and generate meaningful insights from the dataset. These techniques improve the accuracy, efficiency, and usability of the analysis.

Overall, the Job Market Analytics Dashboard Software Hiring Trends in India serves as an effective business intelligence solution for analyzing recruitment trends in the Indian software industry. The dashboard supports data-driven decision-making for job seekers, recruiters, HR professionals, and business analysts by providing clear, interactive, and insightful visual representations of employment data. This project showcases how modern analytics tools can be utilized to solve real-world problems and support strategic career and hiring decisions.

Source of Dataset

The dataset used for developing the Job Market Analytics Dashboard Software Hiring Trends in India was collected from the popular data science platform Kaggle. The dataset is publicly available under the title Indian Job Market Dataset 2025 (97k+ Data Points) and was created by Shivam Shrivastava. It contains more than 97,000 job listings collected from various recruitment platforms across India.

Dataset URL: <https://www.kaggle.com/datasets/shivamshrivastava21/indian-job-market-dataset-2025-2026>

The dataset provides detailed information related to software and technical job postings in India. It includes essential attributes such as job title, company name, location, experience requirements, salary details, skills required, company ratings, and job descriptions. These fields make the dataset highly suitable for recruitment analytics and job market analysis.

The primary objective of using this dataset is to analyze hiring patterns and identify trends in the Indian software job market. Since the dataset contains large-scale recruitment information, it enables meaningful analysis of salary distributions, skill demands, and company hiring activities across multiple cities and industries.

The dataset was specifically selected because of its large volume and structured format. The availability of both categorical and numerical fields allows efficient implementation of data cleaning, transformation, exploratory data analysis, and visualization techniques using Power BI.

Each record in the dataset represents a single job posting collected from online recruitment sources. The dataset includes information such as minimum and maximum salary, minimum and maximum experience, and required technical skills. These features support detailed analytical reporting and dashboard development.

The dataset also includes company-related metrics such as aggregate ratings and employee review counts. These attributes help analyze employer reputation and hiring behavior in the software industry. Such information improves the quality and depth of recruitment analysis performed in the project.

Another important reason for selecting this dataset is its relevance to current industry trends. The dataset reflects modern hiring practices and recruitment demands in India during the years 2025–2026, making the analysis realistic and industry-oriented.

The data was downloaded in CSV format from Kaggle and imported into Power BI for preprocessing and visualization. Before analysis, the dataset underwent multiple preprocessing steps such as handling missing values, formatting salary columns, removing duplicates, and standardizing experience values. These preprocessing steps improved data quality and analytical accuracy.

The dataset contains job postings from different Indian cities such as Bengaluru, Hyderabad, Pune, Chennai, Mumbai, Delhi, and other major IT hubs. This geographical information supports location-based analysis and helps identify cities with high recruitment activity in the software sector.

The job titles available in the dataset include multiple technical roles such as Data Analyst, Software Engineer, Data Scientist, Full Stack Developer, Cloud Engineer, Business Analyst, and Machine Learning Engineer. This diversity allows broad analysis of hiring demand across different software domains.

The dataset also contains skill-related information in the `tagsAndSkills` column. This column includes technologies and programming skills such as Python, SQL, Java, Machine Learning, Cloud Computing, and Data Analytics. These fields are useful for identifying the most in-demand technical skills in the Indian IT market.

The salary information available in the dataset helps perform salary band analysis and compensation trend visualization. Salary ranges are provided for many job postings, allowing comparison between experience levels, locations, and job roles.

The dataset is useful not only for dashboard development but also for machine learning and predictive analytics applications. It can support salary prediction models, job classification systems, skill demand forecasting, and recommendation systems for recruitment analytics.

The dataset creator described the dataset as suitable for exploratory data analysis (EDA), visualization, and machine learning projects. This aligns perfectly with the objectives of the *Job Market Analytics Dashboard* project developed using Power BI.

The dataset is licensed under the *CC BY-NC-SA 4.0* license, which allows users to utilize and analyze the data for educational and research purposes with proper attribution.

Using a real-world recruitment dataset improves the practical relevance of the project. Instead of using synthetic or sample data, this dataset reflects actual hiring trends and recruitment patterns observed in the Indian software industry. This enhances the authenticity of the dashboard analysis.

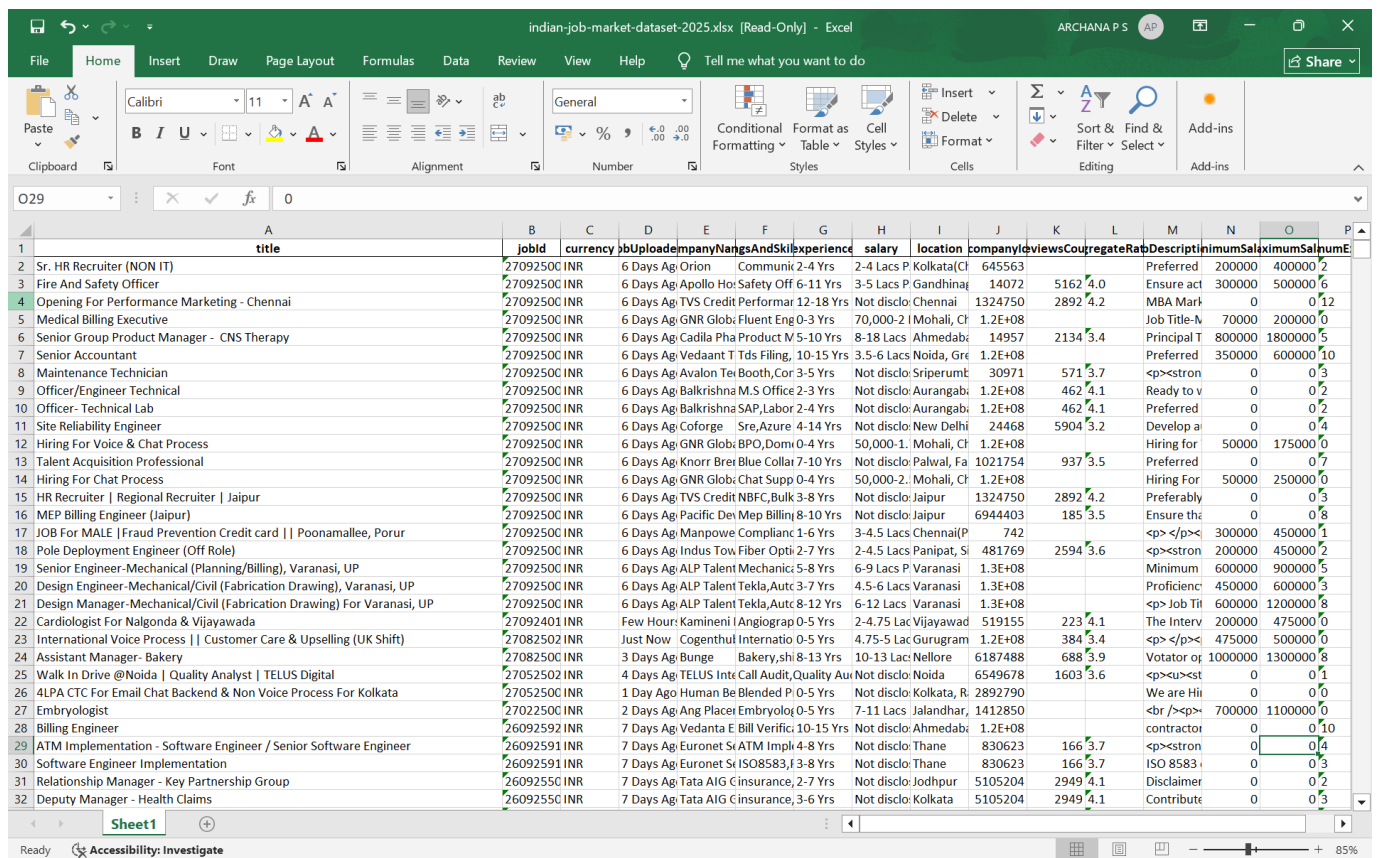
The dataset also supports interactive reporting because of its well-structured attributes and rich feature set. Different visualizations such as KPI cards, treemaps, donut charts, bar charts, and maps can be effectively created using the available columns and metrics.

The source dataset plays a major role in enabling data-driven insights into recruitment trends. By analyzing the collected information, users can understand which companies are hiring actively, which cities have more

opportunities, and which skills are most demanded in the market.

The Kaggle platform itself is widely recognized among data scientists, researchers, and analysts for providing high-quality public datasets and collaborative analytical resources. Using a Kaggle dataset adds credibility and practical significance to the project implementation.

Overall, the Indian Job Market Dataset 2025 (97k+ Data Points) served as the foundational data source for building the Job Market Analytics Dashboard Software Hiring Trends in India. The dataset enabled comprehensive analysis of hiring trends, salary distributions, company openings, experience requirements, and skill demands through interactive Power BI visualizations and business intelligence techniques.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	title	jobid	currency	upload date	company name	location	salary	location	company name	views	cougregate	rap	descripti	minum	salx	minum
1	Sr. HR Recruiter (NON IT)	2709250C	INR	6 Days Ag	Orion	Communic 2-4 Yrs	2-4 Lacs P	Kolkata	645563				Preferred	200000	400000	2
2	Fire And Safety Officer	2709250C	INR	6 Days Ag	Apollo Ho	Safety Off 6-11 Yrs	3-5 Lacs P	Gandhinag	14072	5162	4.0		Ensure act	300000	500000	6
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4	Medical Billing Executive	2709250C	INR	6 Days Ag	GNR Glob	Fluent Eng 0-3 Yrs	70,000-2	Mohali, Ch	1.2E+08				Job Title-N	70000	200000	0
5	Senior Group Product Manager - CNS Therapy	2709250C	INR	6 Days Ag	Cadila Pha	Product M 5-10 Yrs	8-18 Lacs	Ahmedab	14957	2134	3.4		Principal T	800000	1800000	5
6	Senior Accountant	2709250C	INR	6 Days Ag	Vedaant T	Tds Filing, 10-15 Yrs	3.5-6 Lacs	Noida, Gre	1.2E+08				Preferred	350000	600000	10
7	Maintenance Technician	2709250C	INR	6 Days Ag	Avalon Tei	Booth, Cor 3-5 Yrs	Not disclo	Sriperumb	30971	571	3.7		<p><stron	0	0	3
8	Officer/Engineer Technical	2709250C	INR	6 Days Ag	GNR Glob	BPO, Domi 0-4 Yrs	Not disclo	Aurangab	1.2E+08	462	4.1		Ready to v	0	0	2
9	Officer- Technical Lab	2709250C	INR	6 Days Ag	Balkrishna	SAP, Labor 2-4 Yrs	Not disclo	Aurangab	1.2E+08	462	4.1		Preferred	0	0	2
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11	Hiring For Voice & Chat Process	2709250C	INR	6 Days Ag	GNR Glob	BPO, Domi 0-4 Yrs	50,000-1	Mohali, Ch	1.2E+08				Hiring for	50000	175000	0
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18	Senior Engineer-Mechanical (Planning/Billing), Varanasi, UP	2709250C	INR	6 Days Ag	ALP Talent	Mechanic 5-8 Yrs	6-9 Lacs P	Varanasi	1.3E+08				Minimum	600000	900000	5
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20	Design Manager-Mechanical/Civil (Fabrication Drawing) For Varanasi, UP	2709250C	INR	6 Days Ag	ALP Talent	Tekla, Autc 8-12 Yrs	6-12 Lacs	Varanasi	1.3E+08				<p> Job Tit	600000	1200000	8
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28	ATM Implementation - Software Engineer / Senior Software Engineer	2609259I	INR	7 Days Ag	Euronet S	ATM Impl 4-8 Yrs	Not disclo	Thane	830623	166	3.7		<p><stron	0	0	4
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30	Relationship Manager - Key Partnership Group	2609255C	INR	7 Days Ag	Tata AIG	C insurance, 2-7 Yrs	Not disclo	Jodhpur	5105204	2949	4.1		Disclaimer	0	0	2
31	Deputy Manager - Health Claims	2609255C	INR	7 Days Ag	Tata AIG	C insurance, 3-6 Yrs	Not disclo	Kolkata	5105204	2949	4.1		Contribute	0	0	3

Figure 1: Sample view of the Job Market Dataset in Microsoft Excel

Dataset Preprocessing

Data preprocessing is a crucial step in the Job Market Analytics Dashboard — Software Hiring Trends in India project because accurate and meaningful insights depend on clean and well-structured data. The raw dataset collected from Kaggle contained missing values, duplicate records, inconsistent text formatting, and unstructured salary and experience details. Therefore, preprocessing was performed in Power BI using Power Query Editor to transform the raw dataset into an analysis-ready format.

The first step involved importing the CSV dataset into Power BI using the Get Data option. After loading the dataset, the structure of the data was examined carefully by checking column names, data types, null values, and overall dataset quality. Power Query Editor was mainly used for performing data cleaning and transformation operations before visualization.

Handling Missing Values

The dataset contained missing or null values in columns such as salary, company ratings, experience requirements, and skills. These missing values were identified using Power Query tools. Different techniques were applied to handle the missing data effectively.

For numerical columns such as salary and ratings, null values were replaced with average or default values wherever required. For categorical columns such as city, skills, and company names, missing values were either replaced with “Unknown” or removed based on the importance of the records. This process improved the consistency and reliability of the dataset.

Data Cleaning

Data cleaning was performed to remove inconsistencies and improve data quality. Duplicate job postings were identified and removed using the Remove Duplicates feature available in Power Query. This ensured that each job posting was unique and prevented incorrect calculations in dashboard analysis.

Text formatting inconsistencies in columns such as job titles, company names, and city names were corrected using transformation options like Trim, Clean, Capitalize Each Word, and Replace Values. These transformations standardized the dataset and improved filtering, grouping, and visualization accuracy.

Incorrect data types were also corrected during preprocessing. Salary columns were converted into numeric formats, while experience-related columns were transformed into proper number formats. Date columns were converted into Date data types for better time-based analysis and reporting.

Feature Engineering

Feature engineering was performed to create additional analytical columns from the existing dataset. New columns such as salary categories, experience bands, and hiring classifications were created using Custom Columns and DAX calculations.

Salary values were grouped into categories such as Low Salary, Medium Salary, and High Salary ranges.

Similarly, experience levels were categorized into Fresher, Junior-Level, Mid-Level, and Senior-Level groups. These newly created features improved the analytical capability of the dashboard and enabled better comparison across different hiring trends.

Skill-based analysis was also enhanced by extracting frequently mentioned technologies and programming skills such as Python, SQL, Machine Learning, Power BI, and Cloud Computing. This helped identify the most demanded skills in the Indian software industry

Outlier Detection and Treatment

Outlier analysis was performed to identify unrealistic salary values and abnormal experience ranges that could affect dashboard insights. Extremely high or invalid salary records were examined carefully and corrected or removed if necessary. Genuine high-paying job roles were retained because they represent important market trends in the software industry.

Data Transformation

Several transformation techniques were applied in Power Query to prepare the dataset for dashboard visualization. Filtering, sorting, grouping, and aggregation operations were performed to organize the recruitment data effectively.

Data aggregation helped summarize job openings based on cities, companies, salary ranges, and job roles. Conditional columns and calculated measures were also created to support KPI cards, bar charts, treemaps, maps, and donut charts used in the dashboard.

Final Dataset Preparation

After completing all preprocessing steps, the final dataset was validated to ensure consistency, accuracy, and completeness. The cleaned dataset contained properly formatted columns, reduced missing values, and structured recruitment information suitable for visualization in Power BI

The preprocessed dataset became the foundation for developing the Job Market Analytics Dashboard Software Hiring Trends in India. Proper preprocessing improved dashboard performance, visualization accuracy, and analytical reliability, enabling meaningful insights into hiring trends, salary distributions, top companies, and in-demand software skills across India.

Analysis on dataset

The analysis of the Job Market Analysis Dashboard is structured into multiple key objectives, each supported by Power BI visualizations, DAX calculations, and interactive dashboard analysis. The main goal of this project is to understand software hiring trends, salary distributions, company recruitment patterns, experience-level demands, and job opportunities across different cities in India.

1. Job Openings Trend Analysis

Objective

To analyze the overall software job openings and identify recruitment trends across different companies and cities in India

Specific Requirements

- Analyze total number of job postings
- Identify top hiring companies
- Compare job openings across cities
- Calculate total companies and total jobs
- Create:
 - KPI cards for Total Jobs and Total Companies
 - Treemap for Top Companies by Openings
 - Map visualization for Job Openings by City

Analysis Results

- The dashboard shows more than 32K total job openings and 11K hiring companies across India.
- Major IT cities such as Bengaluru, Hyderabad, Chennai, Pune, and Mumbai have high recruitment activity.
- Companies like Tata Consultancy Services, Infosys, Capgemini, Wipro, and Accenture are among the top recruiters
- The software industry shows strong hiring demand across multiple technical domains.

Visualization

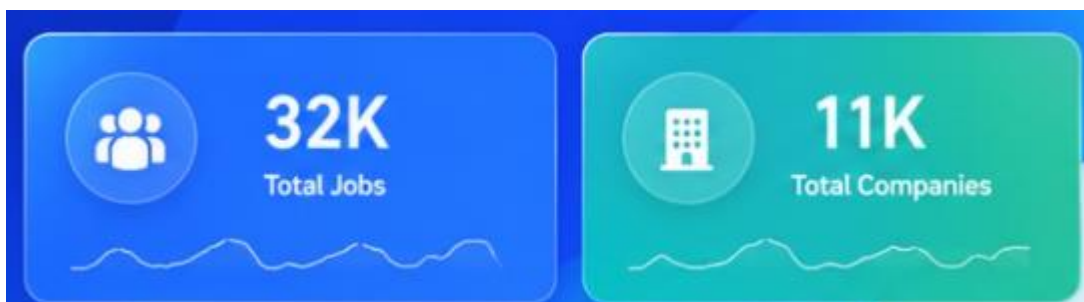


Figure 2: Job Openings Trend Analysis

2. Job Analysis

Objective

To identify the most demanded software job roles in the Indian job market.

Specific Requirements

- Analyze frequency of job roles
- Group job postings by job title
- Identify top technical and non-technical roles
- Create:

Horizontal bar chart for Top Job Roles by Opening

Analysis Results

- Sales Executive, Business Analyst, Data Analyst, Software Engineer, and Java Developer are among the most demanded roles.
- Data-related and software development roles dominate the recruitment market.
- Companies are actively hiring professionals for both technical and business-oriented positions.
- Software engineering and analytics roles show continuous hiring demand

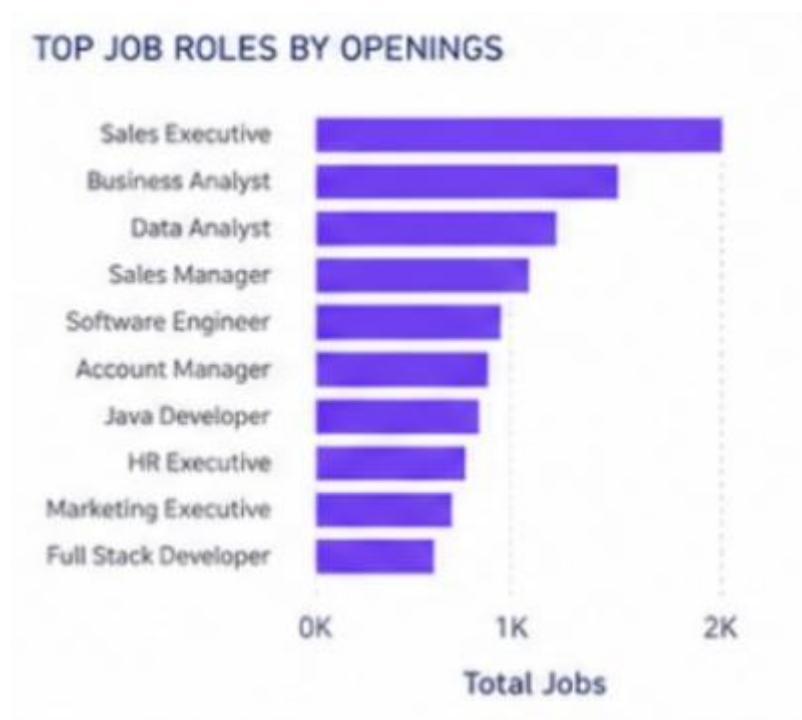


Figure 3: Job Role Analysis

3. Salary Analysis

Objective

To understand salary distribution and compensation trends in the software industry.

Specific Requirements

- Analyze salary bands
 - Compare salary ranges across job roles
 - Calculate average salary
 - Create:

KPI card for Average Salary

Bar chart for Salary Band Distribution

Line chart for Salary vs Experience

Analysis Results

- The average salary shown in the dashboard is approximately 7.61 Lakhs per annum.
- Most job openings fall under the medium salary range categories.
- Senior-level positions offer comparatively higher salary packages.
- Salary increases gradually with experience level and specialization

Visualization

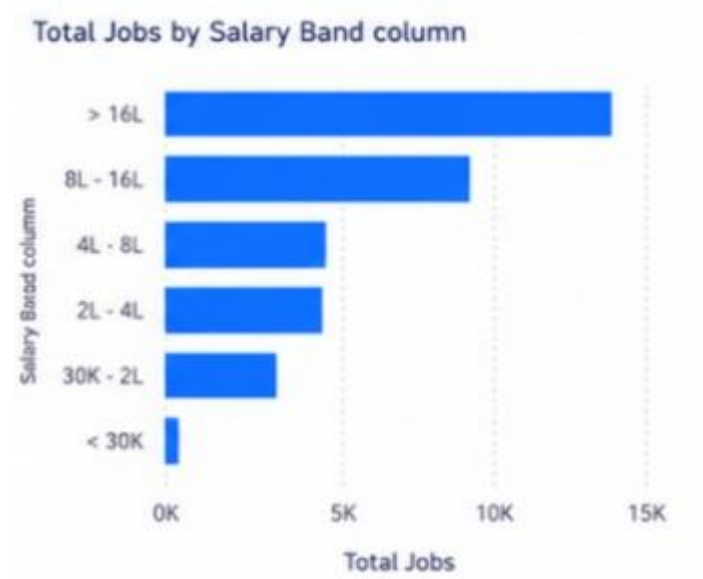


Figure 4: Salary Analysis

4. Experience Level Analysis

Objective

To analyze hiring demand based on candidate experience levels.

Specific Requirements

- Categorize job openings by experience bands
- Compare fresher, junior, mid-level, and senior-level openings
- Create:
 - Donut chart for Experience Band Distribution

Analysis Results

- Junior and Mid-Level positions contribute the highest number of job openings.
- Fresher opportunities are comparatively lower but still present in the market.
- Companies prefer candidates with 1–7 years of experience for most software roles.
- Senior and Lead-level roles are fewer but provide higher salary packages.

Visualization

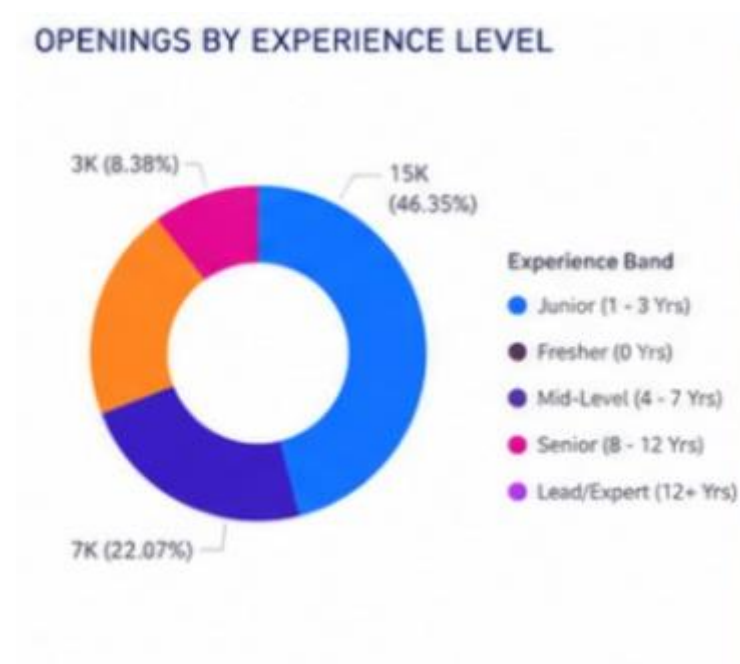


Figure 5: Experience Level Analysis

5. City wise

Analysis Objective

To analyze recruitment activity across different cities and regions.

Specific Requirements

- Group job postings based on city
- Identify cities with maximum hiring activity
- Create:

Map visualization for city-wise job openings

Analysis Results

- Bengaluru has the highest number of software job opportunities.
- Hyderabad, Chennai, Pune, and Mumbai also show strong recruitment demand.
- IT hubs dominate the overall software hiring market in India.
- City-wise analysis helps users identify regions with better employment opportunities

Visualization

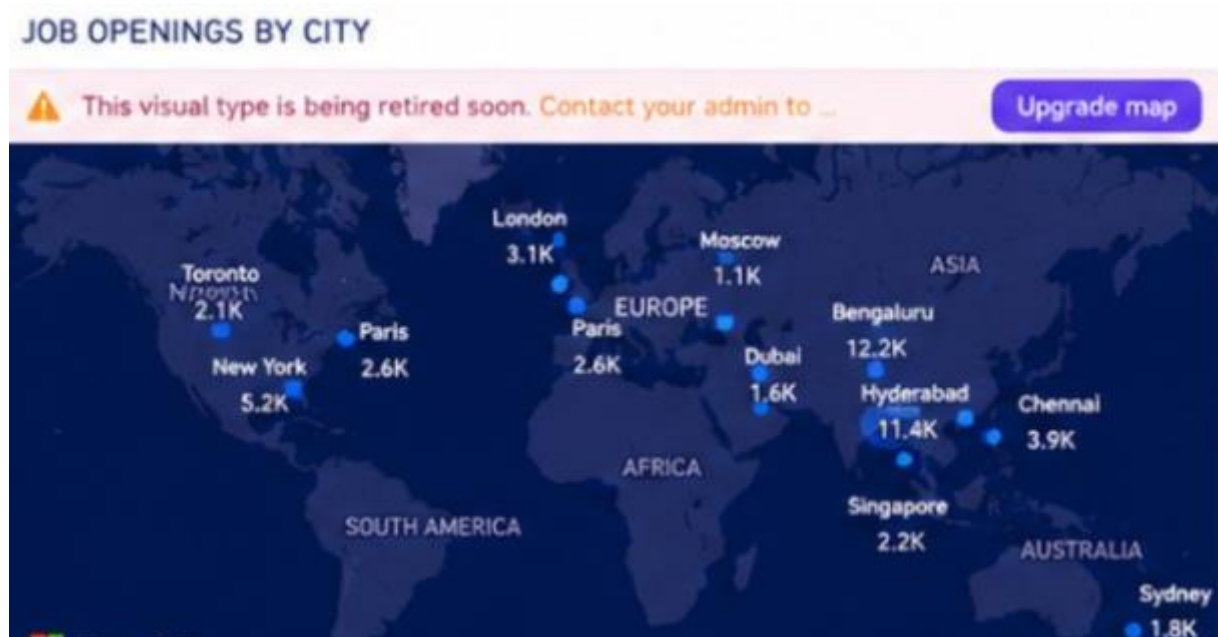


Figure 6: City Wise Recruitment Analysis

6. Hiring Analysis

Objective

To evaluate company recruitment patterns and identify top hiring organizations.

Specific Requirements

- Group dataset by company names
- Calculate:
 - Number of job openings
 - Company hiring frequency
 - Create:
- Treemap visualization

Analysis Results

- Tata Consultancy Services, Infosys, Capgemini, Wipro, and Accenture are major recruiters.
- Large IT service companies dominate the software hiring market.
- Multiple companies are actively recruiting for analytics, cloud, and software development roles.
- Company hiring trends reflect increasing demand for digital transformation technologies.

Visualization



Figure 7: Company Hiring Analysis

7. DI Analysis

Objective

To provide interactive recruitment analysis using Power BI filters and slicers.

Specific Requirements

- Implement slicers for:
 - City
 - Job Title
 - Experience Band
 - Salary Band
 - Enable dynamic filtering across all visuals

Analysis Results

- Interactive slicers improve dashboard usability and analytical flexibility.
- Users can customize recruitment analysis based on their requirements.
- Dynamic filtering helps compare recruitment trends across different categories.
- Power BI interactions improve user experience and support better decision-making.

Visualization

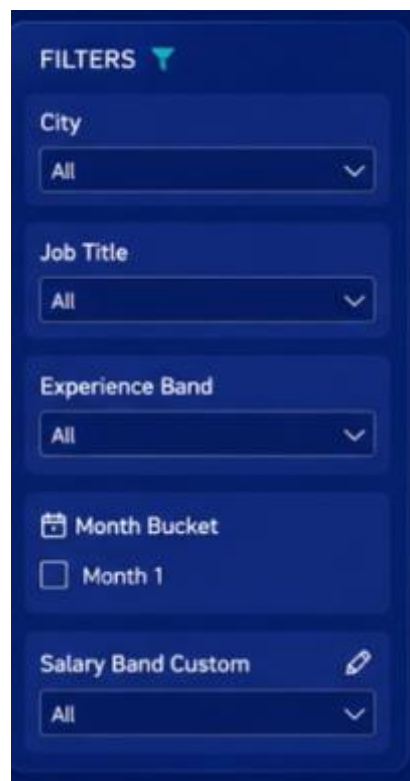


Figure 8: Dashboard Interaction Analysis

Executive Summary

The Job Market Analytics Dashboard Software Hiring Trends in India is developed to analyze and visualize the current hiring trends in the Indian software industry using Power BI. The project focuses on transforming raw recruitment data into meaningful business insights through interactive visualizations and analytical reporting. The dashboard helps users understand employment patterns, salary distributions, top hiring companies, and in-demand technical skills across different regions of India.

The project uses a real-world recruitment dataset collected from Kaggle, which contains more than 97,000 job postings from multiple companies and recruitment platforms. The dataset includes important fields such as job title, company name, city, salary details, experience requirements, company ratings, and technical skills. This large-scale dataset provides a strong foundation for comprehensive recruitment analysis and business intelligence reporting.

Before creating the dashboard, extensive data preprocessing was performed using Power BI Power Query Editor. Data cleaning techniques such as handling missing values, removing duplicate records, correcting data types, trimming unwanted spaces, and standardizing text formats were applied to improve data quality. These preprocessing steps ensured that the dataset became accurate, consistent, and suitable for analysis and visualization.

Feature engineering and data transformation techniques were also implemented to enhance the analytical capabilities of the dashboard. New columns such as salary categories, experience bands, and skill classifications were created to support better segmentation and comparison. Aggregation, filtering, and grouping operations were performed to summarize recruitment trends effectively and generate meaningful insights.

The dashboard was designed using various interactive Power BI visualizations including KPI cards, bar charts, donut charts, maps, and line charts. These visual elements help users easily understand large volumes of recruitment data in a clear and attractive format. Interactive slicers and filters allow users to explore the analysis based on location, company, experience level, salary range, and job role.

The analysis performed in this project identified important hiring trends in major Indian IT cities such as Bengaluru, Hyderabad, Pune, Chennai, Mumbai, and Delhi. The dashboard highlights cities with high recruitment activity and helps users understand regional employment opportunities in the software industry. Geographical visualizations also improve the readability and effectiveness of the analysis.

One of the major findings of the project is the identification of the most demanded technical skills in the current software job market. Skills such as Python, SQL, Machine Learning, Data Analytics, Cloud Computing, and Power BI were found to be highly preferred by companies. This information helps students and professionals understand industry expectations and prepare themselves according to market demand.

The salary analysis included in the dashboard provides insights into compensation trends based on job role, location, and experience level. The dashboard categorizes salary ranges into different bands and helps users compare high-paying and low-paying roles within the software industry. Experience-level analysis also reveals the distribution of fresher, junior-level, mid-level, and senior-level opportunities available in the market.

Overall, the Job Market Analytics Dashboard Software Hiring Trends in India demonstrates the practical application of data analytics, business intelligence, and visualization techniques in solving real-world recruitment analysis problems. The project provides a centralized platform for monitoring hiring trends and supports data-driven decision-making for job seekers, recruiters, HR professionals, and business analysts through interactive and insightful visual representations.

Specific Requirements

The successful execution of the Job Market Analysis Dashboard project requires a well-defined set of requirements that guide the overall workflow, from data collection to final dashboard visualization and interpretation. These requirements can be broadly categorized into functional requirements, non-functional requirements, data requirements, technical requirements, and analytical requirements. Together, these requirements ensure that the dashboard is capable of efficiently processing recruitment data, generating meaningful insights, and presenting them through interactive Power BI visualizations

1. Functional Requirements

The primary functional requirement of the Job Market Analysis Dashboard project is to perform comprehensive analysis on the Indian software job market dataset using Power BI. The system must be capable of importing and handling the dataset in CSV format and performing necessary preprocessing operations such as data cleaning, transformation, feature engineering, and data modeling. The dashboard should allow users to explore different aspects of recruitment data, including job openings, company hiring trends, salary distributions, experience-level demands, and in-demand technical skills.

The dashboard must support the generation of multiple interactive visualizations that help users understand the recruitment trends effectively. These visualizations include KPI cards, bar charts, donut charts, treemaps, line charts, pie charts, and geographical maps. Each visualization should represent a specific analytical objective, such as identifying top hiring companies, analyzing salary trends, understanding experience-level distributions, and examining skill demands across different cities.

Another important functional requirement is the ability to apply filters and slicers for customized analysis. Users should be able to filter the dashboard based on attributes such as city, company name, job role, salary range, and experience level. This enables more focused and interactive exploration of recruitment trends. Additionally, the system should support DAX calculations and measures to compute metrics such as average salary, total job openings, hiring percentages, and skill frequencies.

2. Non-Functional Requirements

Non-functional requirements focus on the performance, usability, reliability, and scalability of the Job Market Analysis Dashboard. The dashboard should efficiently handle large recruitment datasets containing thousands of records without performance delays or system failures. Data loading, transformation, and visualization rendering should execute smoothly within Power BI.

Usability is another critical requirement for the project. The dashboard should have a clean, attractive, and user-friendly interface that allows users to easily navigate between different visualizations and analytical sections. All charts and reports should contain proper labels, titles, legends, and formatting to ensure easy understanding of recruitment trends and analytical outputs.

Reliability and accuracy are also essential non-functional requirements. The system must ensure that the dataset

is processed correctly and that all visualizations accurately represent the underlying recruitment data. Proper preprocessing techniques such as handling missing values, removing duplicates, correcting data types, and standardizing text formats are necessary to maintain data quality and analytical consistency.

3. Data Requirements

The dataset used in the Job Market Analysis Dashboard project must contain comprehensive recruitment information to support meaningful analysis and reporting. Important fields required in the dataset include job title, company name, location, salary details, experience requirements, company ratings, technical skills, and job descriptions. These features allow detailed analysis of software hiring trends and employment patterns in India.

The dataset should be available in a structured format such as CSV to support easy import into Power BI. Before visualization and analysis, the dataset must undergo preprocessing steps including handling missing values, removing duplicate entries, correcting inconsistent formatting, and transforming data types. High-quality and well-structured data is necessary for generating accurate and reliable insights.

Additionally, the dataset should contain a sufficient number of job records collected from different Indian cities and companies. This diversity helps provide a comprehensive overview of the software job market and ensures that the findings represent real-world recruitment trends across India.

4. Technical Requirements

The technical implementation of the Job Market Analysis Dashboard project primarily relies on Power BI and its associated tools for data preprocessing, modeling, and visualization. The main technologies used in the project include:

- **Power BI Desktop** – Used for data visualization, dashboard development, and report generation.
- **Power Query Editor** – Used for data cleaning, transformation, preprocessing, and feature engineering.
- **DAX (Data Analysis Expressions)** – Used for creating calculated columns, measures, KPIs, and advanced analytical computations.
- **CSV Dataset** – Used as the primary data source containing recruitment information.

The system should have sufficient computational resources such as memory and processing power to efficiently load, process, and visualize the recruitment dataset. Proper system performance is necessary to ensure smooth dashboard interaction and fast analytical processing.

5. Analytical Requirements

In addition to technical and functional aspects, the project must satisfy several analytical requirements related to recruitment trend analysis. The dashboard should be capable of:

- Identifying hiring trends across different Indian cities.
- Analyzing salary distributions for various software job roles.
- Examining experience-level demands in the software industry.

- Identifying the most in-demand technical skills such as Python, SQL, Machine Learning, and Cloud Computing.
- Comparing hiring activities across different companies and locations.
- Visualizing recruitment trends using interactive charts and maps.

Each analytical objective should be supported by appropriate Power BI visualizations and DAX calculations. The dashboard should also help users interpret the analytical findings clearly and support data-driven decision-making related to recruitment and career planning.

Analysis Results

The Job Market Analysis Dashboard successfully analyzed recruitment trends in the Indian software industry using interactive Power BI visualizations and business intelligence techniques. The dashboard transformed raw recruitment data into meaningful insights related to job openings, salary trends, experience requirements, and skill demands across different Indian cities and companies.

The analysis revealed that major IT hubs such as Bengaluru, Hyderabad, Pune, Chennai, Mumbai, and Delhi have the highest concentration of software job opportunities. Among these cities, Bengaluru showed the largest number of job openings, indicating its position as the leading technology and recruitment hub in India.

The dashboard identified several top hiring companies actively recruiting software professionals across multiple domains. Companies from IT services, cloud computing, data analytics, and software development sectors contributed significantly to the total number of job postings available in the dataset.

The salary analysis performed in the dashboard highlighted variations in compensation based on job role, experience level, and location. Senior-level positions and specialized technical roles such as Data Scientist, Machine Learning Engineer, and Cloud Engineer offered comparatively higher salary packages than entry-level positions.

Experience-level analysis showed that most companies preferred candidates with mid-level experience ranging between 2 to 5 years. However, a considerable number of fresher-level opportunities were also identified, especially for roles such as Data Analyst, Software Developer, and Support Engineer.

The skill demand analysis revealed that Python, SQL, Machine Learning, Data Analytics, Power BI, and Cloud Computing were among the most frequently required technical skills in the software job market. These technologies appeared consistently across multiple job roles and companies, indicating strong industry demand.

The dashboard also provided insights into hiring distribution based on job categories and technical domains. Software development and data-related roles accounted for the majority of job postings, reflecting the growing importance of data-driven technologies and digital transformation in modern industries.

Interactive filters and slicers implemented in the dashboard improved the analytical flexibility of the project. Users were able to explore recruitment trends based on city, company, experience level, salary range, and technical skills, making the dashboard more user-friendly and suitable for detailed analysis.

The use of Power BI visualizations such as KPI cards, treemaps, bar charts, donut charts, and geographical maps enhanced the clarity and effectiveness of the analysis. The visual storytelling approach made complex recruitment data easier to interpret and supported better understanding of software hiring trends in India.

Overall, the Job Market Analysis Dashboard successfully demonstrated the practical application of data analytics and business intelligence in recruitment trend analysis. The project provided accurate, interactive, and data-driven insights that can support job seekers, recruiters, HR professionals, and business analysts in understanding the evolving software job market in India.

Conclusion

The Job Market Analysis Dashboard successfully demonstrates the use of Power BI and data analytics techniques for analyzing software hiring trends in India. The project converted raw recruitment data into meaningful and visually interactive insights that help users understand employment patterns, company hiring trends, salary distributions, and skill demands in the software industry.

The dashboard analysis revealed that major Indian IT cities such as Bengaluru, Hyderabad, Pune, Chennai, Mumbai, and Delhi have the highest concentration of software job opportunities. Among these cities, Bengaluru emerged as the leading technology hub with the maximum number of recruitment postings and active hiring companies.

The project also identified the most in-demand technical skills required in the software job market. Skills such as Python, SQL, Machine Learning, Data Analytics, Cloud Computing, and Power BI were found to be highly preferred by employers across multiple technical roles. This insight helps students and professionals understand current industry expectations and prepare for future career opportunities.

Salary analysis performed through the dashboard highlighted variations in compensation based on experience level, location, and job role. Senior-level technical positions and specialized roles such as Data Scientist, Machine Learning Engineer, and Cloud Engineer were observed to offer higher salary packages compared to entry-level positions.

The preprocessing and transformation phase played a major role in improving data quality and analytical accuracy. Techniques such as handling missing values, removing duplicate records, correcting data types, and standardizing text fields ensured that the dataset became clean, reliable, and suitable for Power BI visualization and reporting.

Interactive Power BI visualizations such as KPI cards, bar charts, donut charts, treemaps, line charts, and maps enhanced the readability and effectiveness of the analysis. The use of filters and slicers allowed users to customize the dashboard and perform detailed analysis based on city, company, experience level, salary range, and skills.

The project successfully demonstrated how business intelligence tools can be used to solve real-world recruitment analysis problems. By transforming large-scale recruitment data into actionable insights, the dashboard supports data-driven decision-making for job seekers, recruiters, HR professionals, and business analysts.

Overall, the Job Market Analysis Dashboard serves as an effective analytical platform for understanding software hiring trends in India. The project highlights the practical importance of data

analytics, visualization, and Power BI in modern business intelligence applications and provides valuable insights into the evolving Indian software job market

Future Scope

The Job Market Analysis Dashboard can be further enhanced by integrating real-time job market data from online recruitment platforms such as LinkedIn, Naukri, Indeed, and Glassdoor. Real-time data integration would help provide updated hiring trends, live job openings, and current salary insights, making the dashboard more dynamic and industry-relevant.

In the future, advanced machine learning models can be incorporated into the project to perform predictive analytics. These models can help predict future hiring trends, salary growth, skill demand, and emerging job roles in the software industry. Predictive analysis would improve decision-making for both job seekers and recruiters.

The dashboard can also be expanded by including global job market analysis along with Indian recruitment trends. Comparing hiring patterns across different countries and regions would provide broader insights into international employment opportunities, remote work trends, and global skill demands in the software industry.

Another possible enhancement is the implementation of AI-powered recommendation systems. The dashboard could recommend suitable job roles, required skills, certification paths, and career opportunities for users based on their experience, education, and technical background. This would improve personalization and career guidance capabilities.

Future versions of the dashboard can include sentiment analysis on company reviews and employee feedback collected from recruitment platforms. Natural Language Processing (NLP) techniques can be used to analyze employee opinions, workplace culture, and organizational reputation, helping users make informed career decisions.

The project can also be integrated with cloud platforms such as Microsoft Azure, AWS, or Google Cloud for better scalability, real-time processing, and secure data storage. Cloud integration would improve dashboard accessibility, performance, and support large-scale recruitment analytics efficiently.

Additional analytical features such as internship trend analysis, fresher opportunity tracking, remote job analysis, and gender diversity analysis can be incorporated in future development. These features would provide deeper insights into employment patterns and workforce diversity in the software industry.

Overall, the future scope of the Job Market Analysis Dashboard is extensive and can evolve into a complete intelligent recruitment analytics platform. By integrating advanced analytics, artificial intelligence, real-time data processing, and cloud technologies, the project can provide more accurate, interactive, and strategic insights into the continuously evolving software job market.

References

Source Name:

Indian Job Market Dataset 2025 (97k+ Data Points) ([Kaggle](#))

Organization / Author:

Shivam Shrivastava on [Kaggle](#) ([Kaggle](#))

Dataset Link:

[Indian Job Market Dataset 2025-2026](#)

Data Description:

The dataset contains more than 97,000 job listings collected from various recruitment platforms across India. It includes detailed information related to software and technical job postings such as job title, company name, location, salary details, experience requirements, technical skills, company ratings, and job descriptions. Each row in the dataset represents an individual job posting, making it suitable for recruitment analysis, salary trend analysis, skill demand analysis, and business intelligence reporting using Power BI. The dataset is highly useful for data analytics, visualization, machine learning applications, and job market trend analysis projects.